INCH-POUND

MIL-I-25689A(USAF)
23 September 1988
SUPERSEDING
MIL-I-25689(USAF)
13 May 1957

MILITARY SPECIFICATION

INDICATOR, ELECTRICAL TACHOMETER ERU-4/A CRANKSHAFT SPEED, 500-4,000 RPM, 2 INCH SIZE

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of hermetically sealed, remote indicating tachometer indicator, 2 inches in diameter, clamp mounted, and with a range of 500 to 4,000 rpm.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

DD-G-451	Glass, Float Or Plate, Sheet, Figured (Flat, For Glazing, Mirrors And Other Uses)
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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DOD-D-1000	Drawing, Engineering And Associated Lists
MIL-I-5015	Connectors Electrical, Circular Threaded, AN Type, General Specification For
MIL-G-5413	Generator, Tachometer, Four-Pole
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-L-25142	Luminescent Material, Fluorescent
MIL-B-27497	Bearing, Jewel, Sapphire Or Ruby Synthetic
MIL-C-83488	Coating, Aluminum, ION Vapor Deposited

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-838	Lubrication Of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards And Specifications Order Of Preference For The Selection Of
MIL-STD-2073-1A	DOD Material Procedures For Development And Application Of Packaging Requirements

MS STANDARDS

MS3102	Connector, Receptacle, Electric, Box Mounting
MS33558	Numeral And Letter, Aircraft Instrument Dial Standard Form Of
MS33585	Pointer, Dial, Standard Design Of Aircraft Instrument
MS33678	Connector, Receptacle, Electrical Integral Mounting
MS33639	Case, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

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American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.4), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.5.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-970 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts,/and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawing.

3.2.2 Standard parts. With the exception in 3.2.1 MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 MATERIALS

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

3.3.2.1 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be advoided when satisfactory alternative processes can be used.

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3.3.3 Dissimilar metals. Dissimilar metals are defined by MIL-STD-889 and shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing.

3.3.4 Fungus-proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or by protected by hermetic sealing.

3.3.5 Corrosive fumes. The materials, as installed in the indicator, and under the service conditions specified in 3.5, shall not liberate deleterious fumes.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.7 Recycled and reclaimed materials. Recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended use of the item.

3.4 Design and construction. The indicator shall be designed for operation from the varying-frequency alternating-current output of tachometer generators conforming to MIL-G-5413. The frequency of the generator outputs is proportional to the speed at which the generator is rotating. The design shall be such that with increasing and decreasing speeds the pointers shall move smoothly at a uniform rate consistent with the speed change. The indicator shall be constructed so that no parts will work loose in service. The indicator shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The indicator shall be constructed so that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases.

3.4.1 Case. The case shall be in accordance with figure 1 of this specification and shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and features unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 0.031 inch.

3.4.1.1 Case material. The hermetically sealed case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.4.1.2 Finish. The portion of the indicator case exposed to view from the front of the mounting panel shall be finished in durable dull black. The finish applied to the remainder of the case shall be optional, provided that the finish used will in no way prevent compliance with the requirements of this specification.

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3.4.1.3 Filling medium. Where a hermetically sealed can is used the filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (dew point $-65^{\circ}C$) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. If practicable the 100 percent helium filling medium shall be utilized.

3.4.2 Cover glass. The cover glass must meet DD-G-451, type II, quality AA. The thickness of the glass shall be the minimum practicable and entirely suitable for the purpose. Any flaws allowed by the Federal specification shall not interfere with the readability of the instrument.

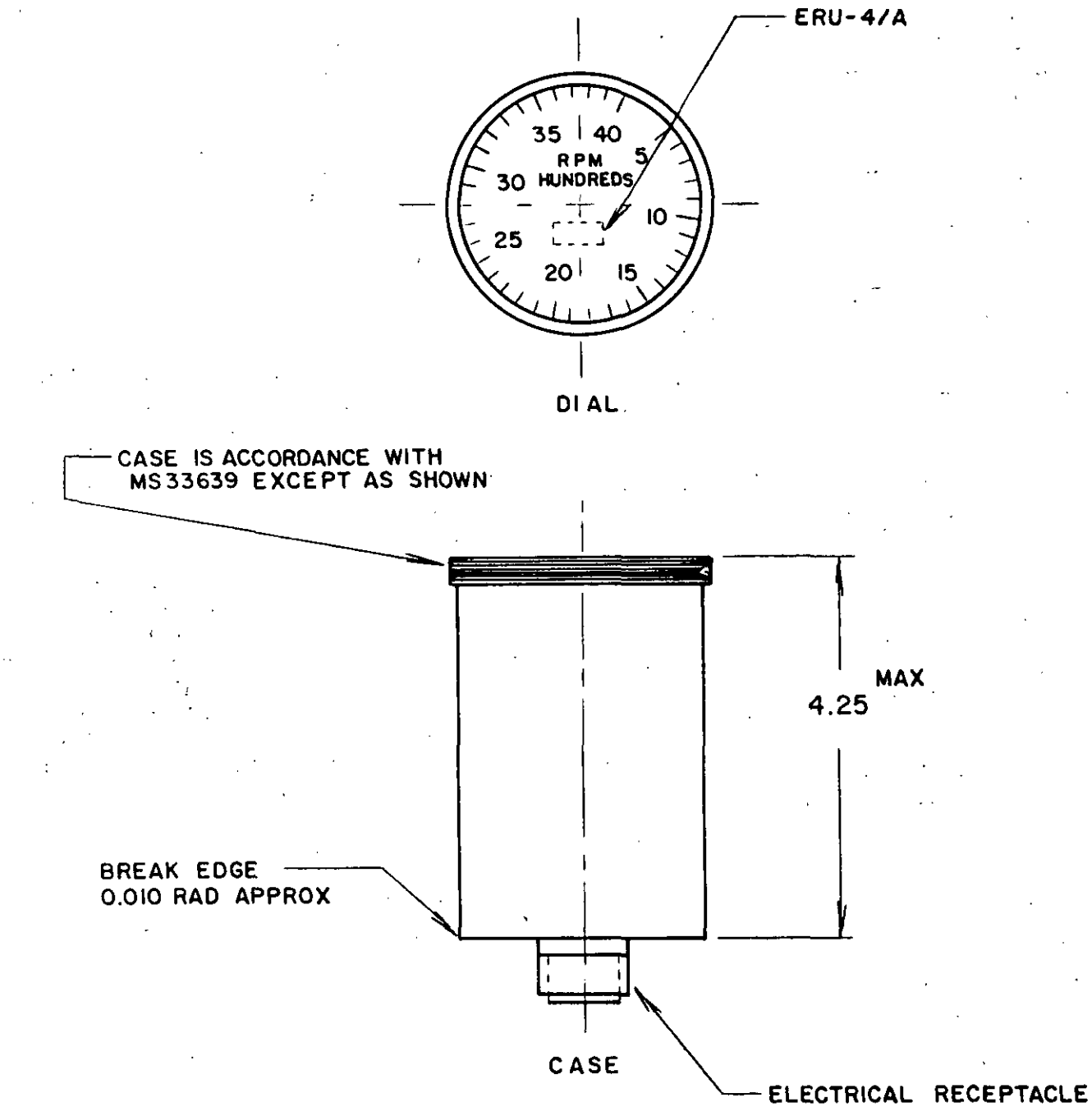
3.4.2.1 Dial to cover glass distance. The distance between the inside surface of the cover glass and the indicator dial shall be the minimum practicable and shall not exceed 0.125 inch.

3.4.3 Dial. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching any dial marking or prevent locating any dial markings in the proper location. The background of the indicator dial and the unshaded portion of the pointer shall be finished in durable dull black. The dial shall have a minimum diameter of 1.75 inches when measured across the outside ends of the graduations.

3.4.3.1 Markings. The dial markings shall conform to figure 1 and shall be durable to withstand usage in service. The 2800 rpm graduation shall be located on the horizontal center line at the left side of the dial. Fluorescent-luminescent marking shall be in accordance with MIL-L-25142, type I or III, as applicable. The style and proportions of numerals and letters placed on the dial shall conform to MS33558, unless otherwise specified. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies shall be cause for rejection. The dial graduations shall be uniformly spaced. The dimensions of the markings shall be as follows:

Markings	Height or Length Inch ± 0.016	Width of Line or Graduation Inch ± 0.005	Material or Finish
Numerals 5, 10, 15, 20, 25, 30, 35 and 40	0.156	0.025	Fluorescent
Graduations at 5, 10 15, 20, 25, 30, 35, and 40	0.156	0.025	Luminescent Material
Lettering "RPM"	0.125	0.016	
Lettering "HUNDREDS"	0.094	0.016	
100-rpm graduations	0.094	0.016	
Shaded portion of pointer			
Lettering "ERU-4/A	0.063	0.010	Durable Dull Black

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DIMENSIONS IN INCHES

FIGURE 1. Outline and dial.

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3.4.4 Electrical receptacle. Each indicator shall be provided with a three-pin electric receptacle conforming to MS33678R10SL-3P or MS3102R10SL-3P, except that it shall be of the hermetically sealed type. The connector shall be located on the vertical and horizontal center lines of the indicator with the polarizing key of the receptacle at the top. The receptacle shall conform to MIL-C-5015, except that the part need not conform to the qualification procedures of that document.

3.4.5 Pointer. The pointer shall conform to MS33585-8, except the length of the pointer shall be such that the tip will extend into the scale a distance equivalent to one-third to two-thirds the length of the shortest graduation.

3.4.6 Indicator. When driven at one-half crankshaft speed, the indicator shall indicate 500 to 4,000 rpm of the engine crankshaft in approximately 330° pointer travel.

3.4.6.1 Scale reading overtravel. The pointer shall be free to rotate beyond the graduation for 4,000 rpm, a distance equivalent to at least 50 rpm.

3.5 Performance. The indicator shall be capable of meeting the tests specified herein when subjected to the following conditions:

a. Room and low temperature starting.-Subjected to starting and synchronous operation at an indicated speed of 100 rpm or greater with the maximum power applied as follows:

1. Room temperature test-0.525V at 150 cycles per minute.

2. -54°C temperature test-0.60V at 150 cycles per minute.

b. Scale error.-Subjected to scale error tests at room temperature, -54°C, -35°C, -5°C and +71°C in accordance with table I.

TABLE I.

Generator Speed RPM	Frequency CPM	Correct Indicated Speed RPM	Scale Error Tolerance				+RPM
			Room Temp	-35°C	-55°C	+71°C	-5°C
250	500	500	30				
300	600	600	30				
400	800	800	30				
500	1000	1000*	30	40	40	40	30
600	1200	1200	30				
700	1400	1400*	30	40	40	40	30
800	1600	1600	30				
900	1800	1800*	30	40	40	40	30
1000	2000	2000*	30	40	50		40
1100	2200	2200*	30	40	50	40	40
1200	2400	2400*	30	50	50	40	40
1300	2600	2600*	30	50	50	50	40
1400	2800	2800*	30	50	60	50	40
1500	3000	3000	30	50	60	50	40
1600	3200	3200*	40	50	60	50	40
1700	3400	3400*	40				
1800	3600	3600*	40	60	60	50	40
1900	3800	3800	40				
2000	4000	4000*	40	60	70	50	40

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* See 4.4.9

c. Friction error. Friction error not to exceed 20 rpm at the 1,000, 1,500, 2,000 and 2,500 rpm test points.

d. Position error. Position error, with the indicator operating at 2,000 rpm, shall not exceed 15 rpm when positioned 45° in any direction from the normal position.

e. Sealing. Case leakage when tested at a pressure differential of 1 atmosphere shall not exceed 0.01 micron cubic foot per hour when the filling medium is nitrogen and helium, and 0.1 micron cubic foot per hour when the filling medium is 100 percent helium.

f. Precipitation and fog. Subjected for a period of 2 hours at an ambient temperature of +85° ±2°C, and them immediately immersed in a liquid bath maintained at 0° ±5°C without evidence of moisture or fog deposit on the inner face of the cover glass.

g. Insulation. Subjected to the potentials specified herein for a period of 5 seconds between the connector pins and any metal part of the case or connector shell without failure.

h. Magnetic effect. Subject the indicator, operating at 1300 rpm and nonoperating, rotated in increments of 45° from normal, when positioned 5.5 inches from the center of a free-magnet compass having a field intensity of 0.17 to 0.19 oersted, without deflecting the compass more than 3°.

i. Vibration. Subjected to vibration error and vibration failure with circular vibration in a plane inclined 45° to the horizontal under the following conditions:

(1) Vibration error not to exceed 30 rpm at an amplitude of 0.009 to 0.011 inch while the frequency is varied from 5 to 50 cycles per second.

(2) Vibration failure, with the indicator operating at 2,000 rpm when subjected to vibration for a 3-hour period at an amplitude between 5 and 50 cycles per second and returned once each hour during the test, with no evidence of failure.

j. Indicator current. The current drawn by each of the 3-phase windings shall not exceed 0.3 amp when connected to a 21V, 4,200 cycle per minute supply.

k. Temperature exposure. Subjected to temperature exposure of -65°C +71°C without failure.

l. Overspeed and reverse operation. Subjected to a generator shaft speed of 2,000 rpm for a period of 5 minutes and then operated in the reverse direction of rotation at a generator shaft speed of 1,000 rpm for a period of 1 minute, without exceeding a scale error of 30 rpm.

m. Endurance. Subjected to 300 hours operation at an indication of 2,600 ±200 rpm without damage,

n. Thermal shock. Subjected to 4 cycles of exposure to water maintained at a temperature of 85° ±2°C and +5° ±2°C, without evidence of leakage when subjected to the sealing test.

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o. Salt fog. Subjected to salt fog for a period of 50 hours in accordance with Procedure I of MIL-STD-810 without evidence of external corrosion or deterioration affecting subsequent operation.

p. Internal examination. Subjected to internal examination for corrosion, deterioration, or damage resulting from all tests specified herein.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part number.

3.7 Wiring. The 3 leads from the indicator windings shall be designed as B, A, and C (conforming to phases 1, 2 and 3). The leads shall be connected to the B, A, and C pins of the electrical receptacle in such a manner that when 3-phase power with phase rotation sequence B-A-C is applied, the resultant indicator reading is positive.

3.7.1 Electrical connections between indicators and generators shall conform to figure 2.

3.8 Stops. The indicator mechanism shall be provided with a stop, which upon reverse operation of the indicator motor will stop the pointer motion after traveling through an arc equal to not less than 6°.

3.9 Lubrication. Lubrication shall be accomplished in accordance with MIL-STD-838, except that the selection of lubricants used shall be approved by the contracting activity.

3.10 Screw threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.11 Weight. The weight of each indicator shall not exceed 1 pound.

3.12 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

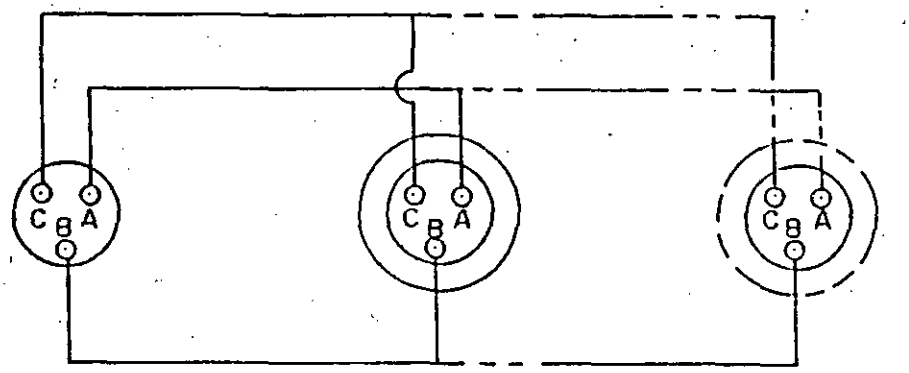
3.13 Workmanship. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner per MIL-STD-454 requirement 9.

3.13.1 General. The indicator, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.13.2 Dimensions. Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.13.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage or injury to the screw or bolt or threads.

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SINGLE TYPE INDICATORS

THE WIRING DIAGRAM SHOWS THE METHOD OF CONNECTIONS WHEN THE GENERATOR DRIVE SHAFT TURNS COUNTERCLOCKWISE (WHEN VIEWED FROM THE SHAFT END). IF THE ROTATION IS CLOCKWISE, REVERSE TWO LEADS AT THE GENERATOR PLUG.

IF TWO INDICATORS ARE USED, THEY SHOULD BE CONNECTED IN PARALLEL WITH THE RESPECTIVE GENERATOR AS SHOWN.

FIGURE 2. Wiring diagram.

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3.13.4 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.13.5 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

3.14 Synthetic bearings. When sapphire or ruby synthetic bearings are used, they shall be in accordance with MIL-B-27497.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).

b. Quality conformance inspection (see 4.2.1).

4.2.1 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests.

b. Sampling plans and tests.

4.2.1.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.4 TEST METHODS of this specification"

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a. Examination of product

b. Room temperature starting

c. Room temperature scale error

d. Friction error

e. Position error

f. Sealing

g. Precipitation or fog

4.2.2 SAMPLING PLANS AND TESTS

4.2.2.1 Sampling plan A. One indicator selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Individual tests

b. Low temperature starting

c. Low temperature warmup

d. Scale error at $-54^{\circ}C$

e. Scale error at $-35^{\circ}C$

f. Scale error at $-5^{\circ}C$

g. Scale error at high temperature

h. Insulation

i. Magnetic effect

j. Vibration error

4.2.2.2 Sampling plan B. Unless otherwise specified, 3 indicators selected at random from the first 15 items produced on the contract or order shall be subjected to the following tests as described under 4.4 TEST METHODS:

a. Sampling plan A tests

b. Indicator current

c. Low Temperature exposure

d. High temperature exposure

e. Vibration failure

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f. Overspeed and reverse operation

g. Endurance

h. Thermal shock

i. Salt fog

j. Internal examination

4.2.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.2.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and methods of correcting them.

4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.3.1 of this specification.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.3.2 Tapping. Except where otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Vibration. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 5 and 50 cycles per second and shall subject the indicator to vibration such that a point on the indicator case will describe in a plane inclined 45° to the horizontal plane, a circle, of the diameter specified herein.

4.3.4 Attitude. Unless otherwise specified, the indicator shall be tested in its normal operating position (dial surface shall be held in vertical plane).

4.3.5 Test voltages. Unless otherwise specified, voltages noted herein are alternating current root-mean-square values. The voltmeter used to measure voltages shall be of a type which is free from frequency errors.

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4.4 TEST METHODS

4.4.1 Examination of product. Each indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, proper operation, and performance.

4.4.2 Room temperature starting. The indicator shall be connected to a variable frequency power supply. The voltage of the supply shall be 0.0035 times the frequency in cycles per minute. The frequency of the supply shall be increased from 0 cycles per minute up to and not exceeding 150 cycles per minute. The indicator may be gently tapped or vibrated and a reading made from the indicator. The reading shall not be less than 100 rpm.

4.4.3 Scale error. The indicator shall be connected, as shown on figure 2, to a generator conforming to MIL-G-5413 and tested at points of the scale indicated in table I. The test shall be made either by driving the generator at a speed to produce the specified indicator readings and reading the speed of the generator or by driving the generator at the specified speeds and reading the indicator. The test shall be made with speeds increasing then with speeds decreasing. With speeds increasing, the speed increasing the speed shall be brought up to, but not exceeding the speed specified to give the desired reading, and with speeds decreasing, the speed shall be brought down to, but shall not fall below the speed specified to give the desire reading.

4.4.3.1 Tolerances test. The errors at the test points shall not exceed the tolerances specified in the applicable table. When the speed is held constant at any point of the scale, the pointer shall not oscillate over a range greater than 20 rpm.

4.4.3.2 Scale error test. In accomplishing the test specified for the scale error test, two single indicators shall be connected to and operated from 1 tachometer generator.

4.4.4 Friction error. The indicator shall be set at the 1,000, 1,500, 2,000, and 2,500 rpm graduations, or at generator speed of 500, 750, 1,000, and 1,200 rpm whichever the contractor finds more convenient. After each setting, the indicator shall be tapped and the change in indications noted. This change shall not exceed 20 rpm.

4.4.5 Position error. The indicator shall be operated at 2,000 rpm with the indicator in the normal (dial vertical) position. The indicator reading shall be noted. The indicator shall then be held in the following positions:

a. Rotated from normal position through 45° to the rear (dial tilted 45° back from vertical position).

b. Rotated from normal position through 45° to the front (dial tilted 45° forward from vertical position).

c. Rotated from normal position through 45° to the right (dial vertical).

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d. Rotated from normal position through 45° to the left (dial vertical).

The maximum deviation from the original reading shall not exceed 15 rpm.

4.4.6 Sealing. The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector. If a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere. If 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 0.1 micron cubic foot per hour.

4.4.7 Low temperature starting. Prior to this test, the room temperature scale error test shall be repeated. The indicator shall then be placed in a chamber, the temperature of which is controllable. The temperature of the circulating air shall be reduced to -54° ±2°C, and held at this temperature for the duration of the test. The indicator shall be kept in the chamber for at least 4 hours after the temperature of -54°C ±2°C is reached before the test is started. No voltage shall be applied to the indicator terminals during this period.

4.4.7.1 Interface. The indicator shall be connected to a variable frequency power supply. The voltage of the supply shall be equal to 0.004 times the frequency in cycles per minute. The frequency of the supply shall be increased from zero cycles per minute up to and not exceeding 150 cycles per minute. The indicator may be gently tapped or vibrated. The indicator reading shall not be less than 150 rpm.

4.4.8 Low temperature warmup. Following the test specified in 4.4.7 the indicator shall be maintained at the -54° ±2°C temperature for a minimum period of 2 hours without voltage applied. The indicator leads shall then be instantaneously connected to a tachometer generator operating at 1,500 rpm. The indicator shall indicate 3,000 ±50 rpm within a period of 5 minutes.

4.4.9 Scale error at -54°C. Following the starting and warmup tests, while still at -54°C, the indicator readings at the points indicated by asterisks in table I shall be determined in the same manner as specified in 4.4.3. The errors at the test points shall not exceed the tolerances specified in the table.

4.4.10 Scale error at -35°C. The indicator readings shall be determined as specified in the scale error test at -54°C, except that the temperature of the air in the chamber shall be -35° ±2°C. The indicator shall be kept at -35° ±2°C at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the table.

4.4.11 Scale error at -5°C. The indicator readings shall be determined as specified in the scale error test at -54°C, except that the temperatures of the air in the chamber shall be -5°C. The indicator shall be kept at -5° ±2°C at least 2 hours prior to the test. The errors at the test points shall not exceed the tolerances specified in the table.

4.4.12 Scale error at high temperature. The indicator scale readings shall be determined as specified in the indicator scale error test at -54°C, except that the temperature of the air in the chamber shall be +71° ±2°C. The indicator shall be kept in the chamber for 4 hours after the temperature of +71 is reached, before the test is started. The errors at the test points shall not exceed the tolerances specified in the table.

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4.4.13 Insulation. Insulation shall be tested by applying a potential, at a commercial frequency, between any pin of the electrical receptacle and any part of the indicator case or the receptacle shall for a minimum period of 5 seconds. In the event the indicator is unsealed the applied voltage shall be 500V. For indicators filled with gas and hermetically sealed, the applied voltage shall be 200V for indicators using the helium filling mixture and 400V for the nitrogen-helium mixture filled indicators. There shall be no breakdown of the insulation as a result of this test.

4.4.14 Magnetic effect. The indicator, first not operating and second operating at an indication of approximately 1,300 rpm, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5.5 inches from and magnetically East or West of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial positions. At each of these positions, the indicators shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.4.15 Vibration error. The indicator reading shall be determined when the generator is operated at a point between 1,600 and 2,000 rpm as selected by the inspector. The indicator shall then be subjected to vibration as specified under test conditions at a diameter of circular motion of 0.009 to 0.011 inch. The indicator readings shall be determined with the generator operating at the original speed as the vibration frequency is varied from 5 to 50 cycles per second. The readings with vibration shall not vary more than 30 rpm from the reading without vibration. The amplitude of the pointer oscillation shall not exceed 30 rpm.

4.4.16 Indicator current. The indicator shall be tested to determine that on a 21V, 4,200-cycle per minute supply, no one of the 3-line currents drawn by the indicator exceeds 0.30 of an amp.

4.4.17 Low temperature exposure. The room temperature scale error test for increasing speeds only shall be repeated. The indicator shall then be subjected to a temperature of $-65^{\circ} \pm 2^{\circ}$C for a period of 48 hours. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements of the scale error test at room temperature, increasing speeds only, and the friction test. There shall be no damage which would adversely affect subsequent operation.

4.4.18 High temperature exposure. The indicator shall be placed within a chamber and the internal temperature of the chamber raised to $+71^{\circ} \pm 2^{\circ}$C with an internal relative humidity of not more than 5 percent. The source of heat for the chamber shall be arranged in such a manner that radiant heat shall not fall upon the test specimen. The total volume occupied by a single item of equipment or by several items shall not exceed 50 percent of the internal volume of the test chamber. The indicator shall be maintained at $+71^{\circ}$ $2^{\circ} \pm$C for a period of 24 hours. While still at this temperature, the indicator shall be operated throughout its range, for increasing speeds only, without exhibiting sticking or erratic characteristics. After the chamber has returned to room temperature and the indicator has remained at room temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements of the scale error at room temperature test, increasing speeds only, and the friction test. There shall be no damage which would adversely affect subsequent operation.

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4.4.19 Vibration failure. The indicator shall be mounted on the vibration stand in its normal operating position. The test shall be conducted while the indicator is being operated at an indication of approximately 2,000 rpm. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 inch for a 3-hour period. The frequency shall be varied uniformly from 5 to 50 cycles per second and return once each hour during this test. After the 3-hour period, the indicator shall be subjected to the scale error at room temperature test, increasing speeds only. The readings shall not differ from the reading before vibration by more than 30 rpm. In addition, no looseness in the mechanism or damage to any part of the instrument shall result from this test.

4.4.20 Overspeed and reverse operation. One or two of the indicator mechanisms shall be electrically connected to a tachometer generator. The reading of the indicators shall be determined when operated at 1,000 and 3,000 rpm. The indicators shall be at room temperature and the readings shall be taken before the indicators have attained an appreciable temperature rise.

4.4.20.1 Generator and indicators. The generator and indicators shall then be operated first at a generator shaft speed of approximately 2,200 rpm for a period of 5 minutes, and then in the reverse direction of rotation at a generator drive shaft speed of approximately 1,000 rpm for a period of 1 minute.

4.4.20.2 Indicated speed. After the indicators have been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 1,000 and 3,000 rpm. The change in indicated speed at the respective scale points before and after the overspeed test shall not be more than 30 rpm.

4.4.21 Endurance. The readings of the indicator shall be determined when operated at 1,000 and 3,000 rpm. The indicator shall be at room temperature and the readings shall be taken before the indicator has attained an appreciable temperature rise. The indicator shall then be operated at an indication of 2,600 $\pm$200 rpm for a period of 300 hours. The voltage shall be measured at the indicator terminals and shall be 18.5 $\pm$1V during the 300 hour period. At the end of the 300 hour period and after the indicator has been allowed to cool to room temperature, the readings of the indicator shall again be determined when operated at 1,000 and 3,000 rpm. The change in reading obtained before and after this test shall not be more than 30 rpm. No screws or other parts shall become loose or damaged as a result of this test. Indicators which have been subjected to all the tests of this specification up to and including this test shall also be subjected to the scale error at room temperature, friction and position error tests. Scale error shall not be 20 rpm more than allowed at room temperature in table I. Friction error shall not exceed 20 rpm. Position error shall not exceed 20 rpm.

4.4.22 Thermal shock. The indicator shall be subjected to four cycles of exposure to water at 71° $\pm$2°C and 5° $\pm$2°C without evidence of moisture penetration or damage to coating or enclosures. Each cycle of the test shall consist of immersing the indicator in water at 71°C for a period of 30 minutes and then within 5 seconds after removal from the bath, the indicator shall be immersed for a period of 30 minutes in the other bath maintained at 5°C. This cycle shall be repeated continuously, one cycle following the other, until 4 cycles have been completed. Following this test, the indicator shall be subjected to the sealing test. No indicator leakage shall occur as a result of this test.

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4.4.23 Precipitation or fog. The indicator shall be placed in a chamber, the air temperature of which is controllable. The temperature of this circulating air shall be -71° $\pm 2^{\circ}$C and the indicator shall be kept in this chamber for a minimum of 2 hours. The indicator shall then be removed from the high temperature chamber and within 30 seconds the indicator face shall be immersed to a depth of 1 to 2 inches in a fluid bath maintained at 0° to 5°C. After 5 minutes, the indicator shall be removed from the fluid bath and dried. There shall be no evidence of moisture or other deposits on the inner face of the cover glass.

4.4.24 Salt fog. The indicator shall be subjected to a salt fog test in accordance with Procedure I of MIL-STD-810 for a period of 50 hours. External connections shall be made to the indicator in such a manner as to simulate installed conditions. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation. The indicator shall then be subjected to the insulation and leak tests as specified herein. The external surface of the indicator may be washed with distilled water and air dried prior to conduction of the insulation test.

4.4.25 Internal examination. The indicator shall be examined internally for evidence of corrosion or deterioration resulting from the test specified herein. There shall be no evidence of corrosion or deterioration.

4.5 FIRST ARTICLE INSPECTION

4.5.1 First article inspection samples. The first article inspection samples shall consist of four indicators. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization. The first article inspection consist of all tests specified under 4.4 Test Methods.

4.5.2 First article required. Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualifications tests.

4.6 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073/1A, as specified (see 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservative shall not be used.

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5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure indicator does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tachometer indicator covered by this specification is intended for use to indicate the crankshaft speed of aircraft reciprocating engines.

6.2 Ordering data. Procurement documents should specify the following:

a. Whether sampling plan B is to be omitted.

b. Selection of applicable levels of packaging and packing (see Section 5).

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6.3 Definitions.

6.3.1 Tapping. The phrase "gently tapped" is defined as meaning the application of light vibration sufficient to overcome the quality aircraft instruments.

6.3.2 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4 First article. When first article inspection is required, the contracting officer should provide specifc guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first article production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.5. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

ERU4/A
Indicator
MIL-G-5413
Tachometer
500-4000

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-25689A	INDICATOR, ELECTRICAL TACHOMETER ERU-4/A CRANKSHAFT SPEED, 500-4,000 RPM, 2 INCH SIZE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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